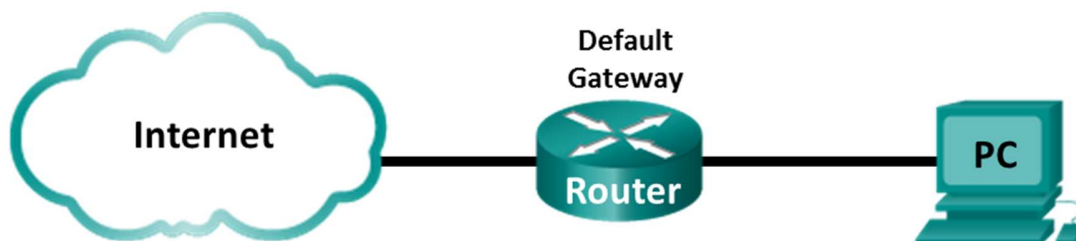


Week 5 Tutorial 2 – Using Wireshark to Examine Ethernet Frames

Topology



Objectives

Part 1: Examine the Header Fields in an Ethernet II Frame

Part 2: Use Wireshark to Capture and Analyse Ethernet Frames

Background / Scenario

When upper layer protocols communicate with each other, data flows down the Open Systems Interconnection (OSI) layers and is encapsulated into a Layer 2 frame. The frame composition is dependent on the media access type. For example, if the upper layer protocols are TCP and IP and the media access is Ethernet, then the Layer 2 frame encapsulation will be Ethernet II. This is typical for a LAN environment.

When learning about Layer 2 concepts, it is helpful to analyse frame header information. In the first part of this lab, you will review the fields contained in an Ethernet II frame. In Part 2, you will use Wireshark to capture and analyse Ethernet II frame header fields for local and remote traffic.

Required Resources

- 1 PC (Windows 7, 8 or 10 with Internet access and Wireshark installed)

Part 1: Examine the Header Fields in an Ethernet II Frame

In Part 1, you will examine the header fields and content in an Ethernet II Frame. A Wireshark capture will be used to examine the contents in those fields.

Step 1: Review the Ethernet II header field descriptions and lengths.

Preamble	Destination Address	Source Address	Frame Type	Data	FCS
8 Bytes	6 Bytes	6 Bytes	2 Bytes	46 – 1500 Bytes	4 Bytes

Using Wireshark to Examine Ethernet Frames

Step 2: Examine the network configuration of the PC.

This PC host IP address is 192.168.1.17 and the default gateway has an IP address of 192.168.1.1.

```
Wireless LAN adapter Wireless Network Connection:
Connection-specific DNS Suffix . : 
Description . . . . . : Broadcom 802.11a/b/g WLAN
Physical Address. . . . . : 00-1A-73-EA-63-8C
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::a858:5f3e:35e2:d38f%13(Preferred)
IPv4 Address. . . . . : 192.168.1.17(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Tuesday, June 16, 2015 6:59:54 AM
Lease Expires . . . . . : Wednesday, June 17, 2015 6:59:54 AM
Default Gateway . . . . . : 192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 234887795
DHCPv6 Client DUID. . . . . : 00-01-00-01-1B-07-0A-E1-00-1E-EC-15-74-C2

DNS Servers . . . . . : 192.168.1.1
NetBIOS over Tcpip. . . . . : Enabled
```

Step 3: Examine Ethernet frames in a Wireshark capture.

The Wireshark capture below shows the packets generated by a ping being issued from a PC host to its default gateway. A filter has been applied to Wireshark to view the ARP and ICMP protocols only. The session begins with an ARP query for the MAC address of the gateway router, followed by four ping requests and replies.

The image shows a Wireshark network capture with a filter of 'arp or icmp'. The packet list shows five packets: an ARP request (No. 9), an ARP reply (No. 10), and three ping requests (Nos. 11, 12, 14) and one ping reply (No. 15). The packet details for Frame 9 are expanded, showing the Ethernet II header (Source: GemtekTe_ea:63:8c, Destination: Broadcast) and the ARP (request) header. The packet bytes are displayed in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
9	2.497611000	GemtekTe_ea:63:8c	Broadcast	ARP	42	who has 192.168.1.1? Tell 192.168.1.17
10	2.502719000	Netgear_ea:b1:7a	GemtekTe_ea:63:8c	ARP	42	192.168.1.1 is at 80:37:73:ea:b1:7a
11	2.502767000	192.168.1.17	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=19/4864,
12	2.503610000	192.168.1.1	192.168.1.17	ICMP	74	Echo (ping) reply id=0x0001, seq=19/4864,
14	3.499098000	192.168.1.17	192.168.1.1	ICMP	74	Echo (ping) request id=0x0001, seq=20/5120,
15	3.501917000	192.168.1.1	192.168.1.17	ICMP	74	Echo (ping) reply id=0x0001, seq=20/5120,

Frame 9: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 0

Ethernet II, Src: GemtekTe_ea:63:8c (00:1a:73:ea:63:8c), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Destination: Broadcast (ff:ff:ff:ff:ff:ff)

Source: GemtekTe_ea:63:8c (00:1a:73:ea:63:8c)

Type: ARP (0x0806)

Address Resolution Protocol (request)

Offset	Hex	ASCII
0000	ff ff ff ff ff ff 00 1a 73 ea 63 8c 08 06 00 01	S.C.....
0010	08 00 06 04 00 01 00 1a 73 ea 63 8c c0 a8 01 11	S.C.....
0020	00 00 00 00 00 00 c0 a8 01 01	

Using Wireshark to Examine Ethernet Frames

Step 4: Examine the Ethernet II header contents of an ARP request.

The following table takes the first frame in the Wireshark capture and displays the data in the Ethernet II header fields.

Field	Value	Description						
Preamble	Not shown in capture	This field contains synchronizing bits, processed by the NIC hardware.						
Destination Address	Broadcast (ff:ff:ff:ff:ff:ff)	Layer 2 addresses for the frame. Each address is 48 bits long, or 6 octets, expressed as 12 hexadecimal digits, 0–9, A–F. A common format is 12:34:56:78:9A:BC. The first six hex numbers indicate the manufacturer of the network interface card (NIC), the last six hex numbers are the serial number of the NIC. The destination address may be a broadcast, which contains all ones, or a unicast. The source address is always unicast.						
Source Address	GemtekTe_ea:63:8c (00:1a:73:ea:63:8c)							
Frame Type	0x0806	For Ethernet II frames, this field contains a hexadecimal value that is used to indicate the type of upper-layer protocol in the data field. There are numerous upper-layer protocols supported by Ethernet II. Two common frame types are: <table><tr><th>Value</th><th>Description</th></tr><tr><td>0x0800</td><td>IPv4 Protocol</td></tr><tr><td>0x0806</td><td>Address resolution protocol (ARP)</td></tr></table>	Value	Description	0x0800	IPv4 Protocol	0x0806	Address resolution protocol (ARP)
Value	Description							
0x0800	IPv4 Protocol							
0x0806	Address resolution protocol (ARP)							
Data	ARP	Contains the encapsulated upper-level protocol. The data field is between 46 – 1,500 bytes.						
FCS	Not shown in capture	Frame Check Sequence, used by the NIC to identify errors during transmission. The value is computed by the sending machine, encompassing frame addresses, type, and data field. It is verified by the receiver.						

What is significant about the contents of the destination address field?

Why does the PC send out a broadcast ARP prior to sending the first ping request?

What is the MAC address of the source in the first frame? _____

What is the Vendor ID (OUI) of the Source's NIC? _____

What portion of the MAC address is the OUI?

What is the Source's NIC serial number? _____

Part 2: Use Wireshark to Capture and Analyse Ethernet Frames

In Part 2, you will use Wireshark to capture local and remote Ethernet frames. You will then examine the information that is contained in the frame header fields.

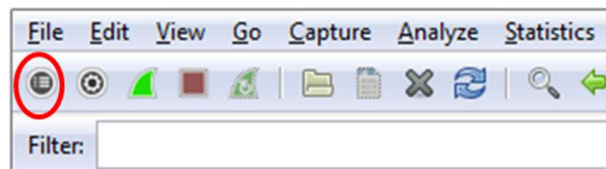
Step 1: Determine the IP address of the default gateway on your PC.

Open a command prompt window and issue the **ipconfig** command.

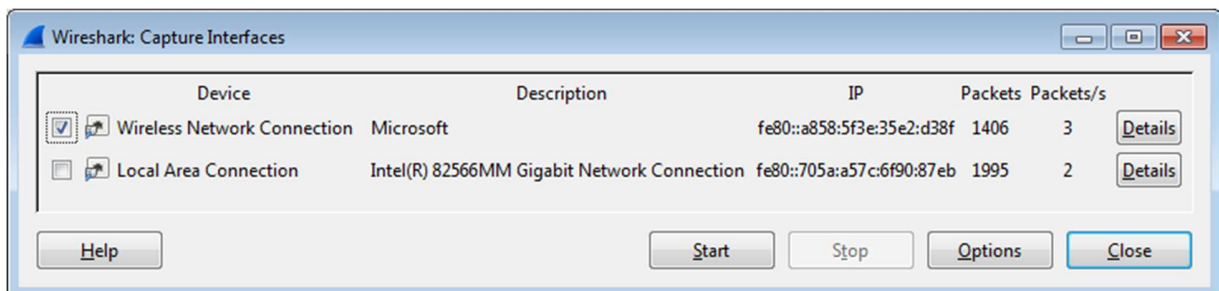
What is the IP Address of the PC Default Gateway? _____

Step 2: Start capturing traffic on your PC's NIC.

- Open Wireshark.
- On the Wireshark Network Analyser toolbar, click the **Interface List** icon.



- On the Wireshark: Capture Interfaces window, select the interface to start traffic capturing by clicking the appropriate check box, and then click **Start**. If you are uncertain of what interface to check, click **Details** for more information about each interface listed.



- Observe the traffic that appears in the Packet List window.

A screenshot of the Wireshark Packet List window. It shows a list of captured packets with columns for No., Time, Source, Destination, Protocol, Length, and Info. The list includes various protocols such as DNS, ARP, ICMPv6, and IGMPv3.

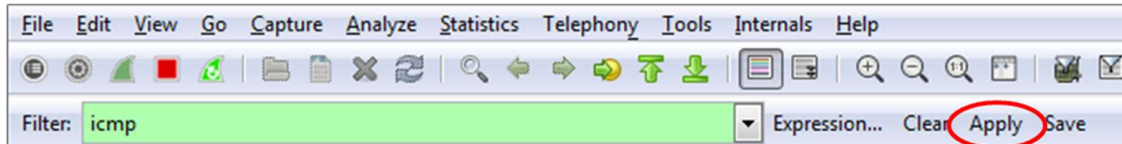
No.	Time	Source	Destination	Protocol	Length	Info
17	3.691404000	192.168.1.17	192.168.1.1	DNS	85	Standard query 0x0c33 A teredo.ipv6.microso
18	3.702954000	192.168.1.1	192.168.1.17	DNS	150	Standard query response 0x0c33 CNAME teredo
19	3.752602000	GemtekTe_ea:63:8	Broadcast	ARP	42	who has 192.168.1.1? Tell 192.168.1.17
20	3.754732000	Netgear_ea:b1:7	GemtekTe_ea:63	ARP	42	192.168.1.1 is at 80:37:73:ea:b1:7a
21	3.768583000	fe80::a858:5f3e:ff02::16		ICMPv6	90	Multicast Listener Report Message v2
22	3.768843000	192.168.1.17	224.0.0.22	IGMPv3	54	Membership Report / Leave group 224.0.0.252
23	3.795917000	GemtekTe_ea:63:8	Broadcast	ARP	42	who has 192.168.1.1? Tell 192.168.1.17
24	3.800804000	Netgear_ea:b1:7	GemtekTe_ea:63	ARP	42	192.168.1.1 is at 80:37:73:ea:b1:7a

Using Wireshark to Examine Ethernet Frames

Step 3: Filter Wireshark to display only ICMP traffic.

You can use the filter in Wireshark to block visibility of unwanted traffic. The filter does not block the capture of unwanted data; it only filters what to display on the screen. For now, only ICMP traffic is to be displayed.

In the Wireshark **Filter** box, type **icmp**. The box should turn green if you typed the filter correctly. If the box is green, click **Apply** to apply the filter.

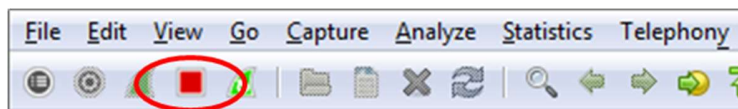


Step 4: From the command prompt window, ping the default gateway of your PC.

From the command window, ping the default gateway using the IP address that you recorded in Step 1.

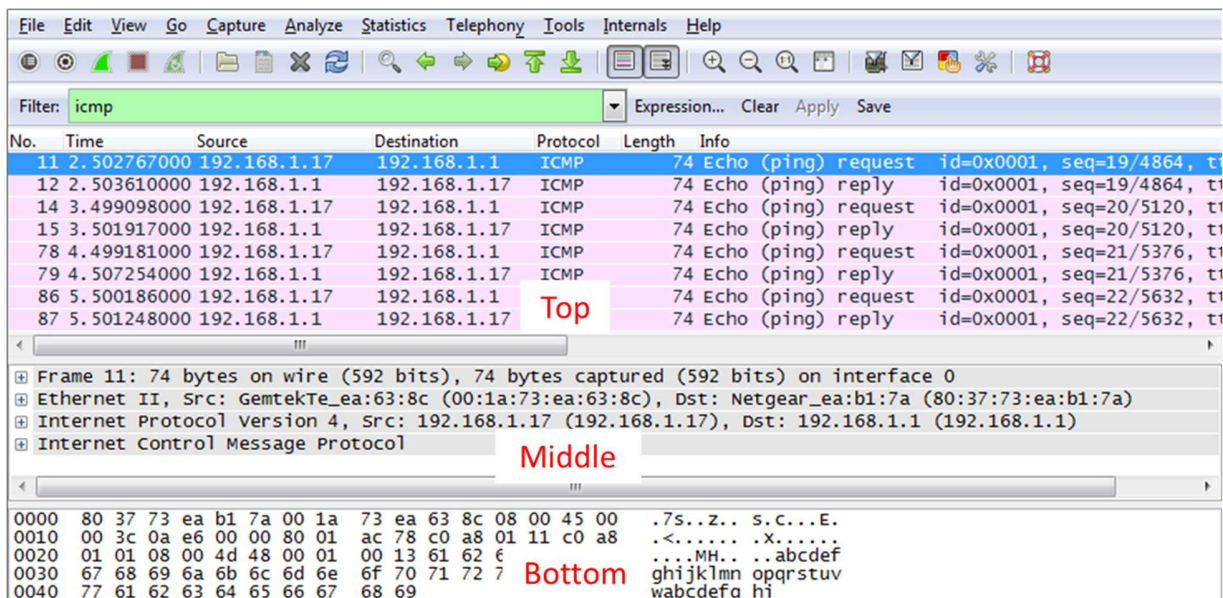
Step 5: Stop capturing traffic on the NIC.

Click the **Stop Capture** icon to stop capturing traffic.



Step 6: Examine the first Echo (ping) request in Wireshark.

The Wireshark main window is divided into three sections: the Packet List pane (top), the Packet Details pane (middle), and the Packet Bytes pane (bottom). If you selected the correct interface for packet capturing in Step 3, Wireshark should display the ICMP information in the Packet List pane of Wireshark, similar to the following example.



- In the Packet List pane (top section), click the first frame listed. You should see **Echo (ping) request** under the **Info** heading. This should highlight the line blue.

Using Wireshark to Examine Ethernet Frames

- b. Examine the first line in the Packet Details pane (middle section). This line displays the length of the frame; 74 bytes in this example.
- c. The second line in the Packet Details pane shows that it is an Ethernet II frame. The source and destination MAC addresses are also displayed.

What is the MAC address of the PC's NIC? _____

What is the default gateway's MAC address? _____

- d. You can click the plus (+) sign at the beginning of the second line to obtain more information about the Ethernet II frame. Notice that the plus sign changes to a minus (-) sign.

What type of frame is displayed? _____

- e. The last two lines displayed in the middle section provide information about the data field of the frame. Notice that the data contains the source and destination IPv4 address information.

What is the source IP address? _____

What is the destination IP address? _____

- f. You can click any line in the middle section to highlight that part of the frame (hex and ASCII) in the Packet Bytes pane (bottom section). Click the **Internet Control Message Protocol** line in the middle section and examine what is highlighted in the Packet Bytes pane.

+ Frame 11: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0		
- Ethernet II, Src: GemtekTe_ea:63:8c (00:1a:73:ea:63:8c), Dst: Netgear_ea:b1:7a (80:37:73:ea:b1:7a)		
+ Destination: Netgear_ea:b1:7a (80:37:73:ea:b1:7a)		
+ Source: GemtekTe_ea:63:8c (00:1a:73:ea:63:8c)		
Type: IP (0x0800)		
+ Internet Protocol Version 4, Src: 192.168.1.17 (192.168.1.17), Dst: 192.168.1.1 (192.168.1.1)		
- Internet Control Message Protocol		
Type: 8 (Echo (ping) request)		
Code: 0		
Checksum: 0x4d48 [correct]		
!!!		
0000	80 37 73 ea b1 7a 00 1a 73 ea 63 8c 08 00 45 00	.7s..Z.. s.C...E.
0010	00 3c 0a e6 00 00 80 01 ac 78 c0 a8 01 11 c0 a8	.<..... .X.....
0020	01 01 08 00 4d 48 00 01 00 13 61 62 63 64 65 66	.. .MH.. ..abcdef
0030	67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76	ghijklmn opqrstuv
0040	77 61 62 63 64 65 66 67 68 69	wabdefg hi

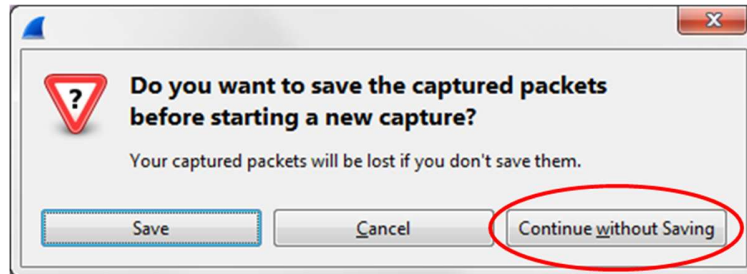
What do the last two highlighted octets spell? _____

- g. Click the next frame in the top section and examine an Echo reply frame. Notice that the source and destination MAC addresses have reversed, because this frame was sent from the default gateway router as a reply to the first ping.

What device and MAC address is displayed as the destination address?

Step 7: Restart packet capture in Wireshark.

Click the **Start Capture** icon to start a new Wireshark capture. You will receive a popup window asking if you would like to save the previous captured packets to a file before starting a new capture. Click **Continue without Saving**.



Step 8: In the command prompt window, ping www.cisco.com.

Step 9: Stop capturing packets.

Step 10: Examine the new data in the packet list pane of Wireshark.

In the first echo (ping) request frame, what are the source and destination MAC addresses?

Source: _____

Destination: _____

What are the source and destination IP addresses contained in the data field of the frame?

Source: _____

Destination: _____

Compare these addresses to the addresses you received in Step 6. The only address that changed is the destination IP address. Why has the destination IP address changed, while the destination MAC address remained the same?

Reflection

Wireshark does not display the preamble field of a frame header. What does the preamble contain?
